CoExp Web Application, a web-based tool for the interactive use of gene co-expression networks

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Background

It is challenging to explore and manage gene co-expression networks (GCN) from a command-line environment.

Methods

CoExp Web Application has been developed to facilitate the exploitation of GCN through a web-based interface. The front-end has been built following a Model View Controller (MVC) architecture implemented through the ASP.NET Core MVC framework. The back end is based on the family of CoExpNets R package. The bridge between both ends has been built using a REST API published by the Plumber R package.

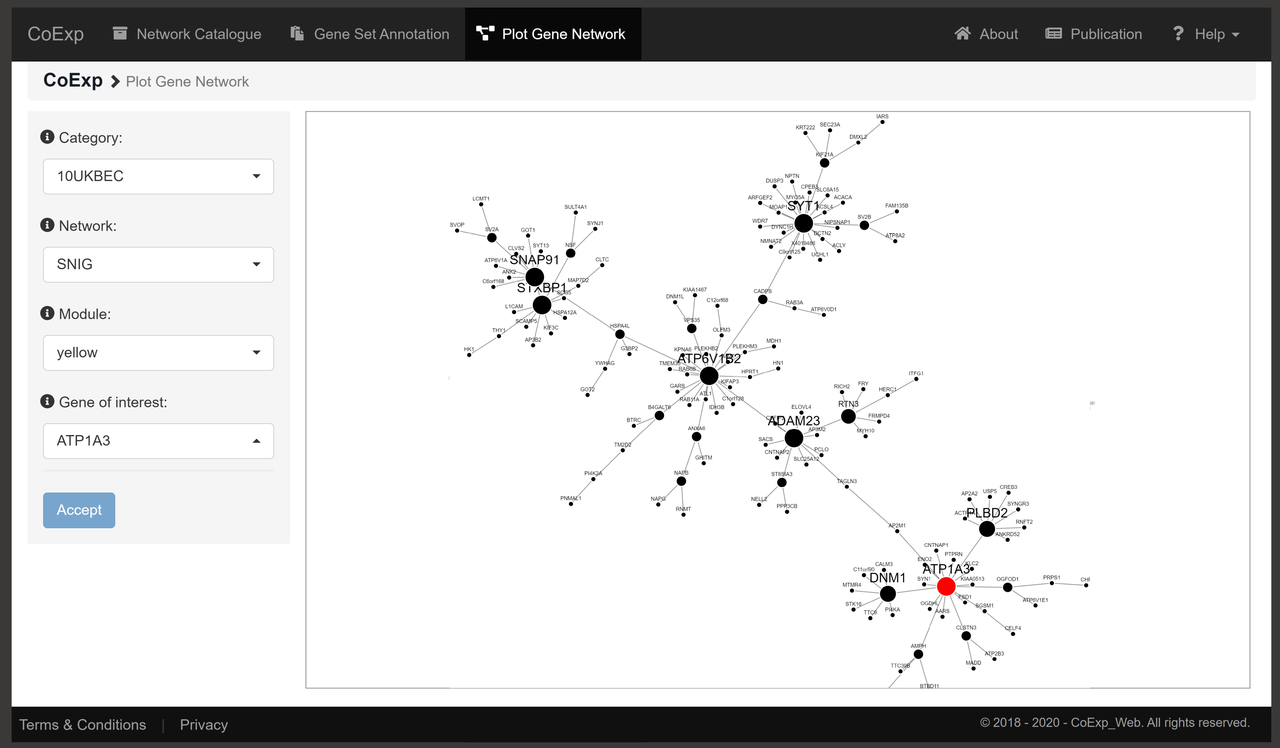
Results

CoExp Web Application allows the exploitation of 109 networks by (1) gene ontology and cell type; (2) through the annotation of a list of genes, and (3) by the generation of interactive graphs. CoExp Web Application can be used from a wide range of devices enabling web browsing as well as published through the Docker container technology.

Conclusion

CoExp Web Application makes gene co-expression models visually manageable, accessible and easily exploitable to the scientific community.

CoExp Web Application offers the **visual and interactive** exploitation of **109 co-expression** networks.



CoExp Web Application allows, among other options, the **generation of an undirected graph** to represent the most **important genes neighbouring a gene of interest** (in red).

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Links

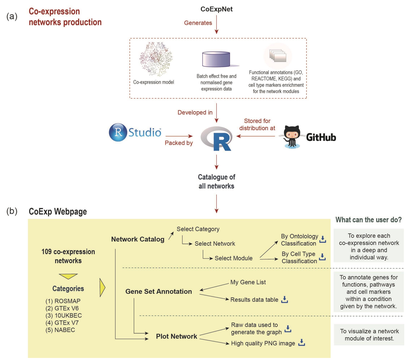
https://rytenlab.com/coexp/

https://github.com/SoniaRuiz

https://github.com/juanbot

https://hub.docker.com/r/soniaruiz/coexp

Life cycle of CoExp



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